

SECTION 17000
INSTRUMENTATION & CONTROL SYSTEM (ICS) GENERAL
INFORMATION

PART 1 - GENERAL

1.1 SUMMARY

- A. The following is a detailed description of Engineering services to be provided by the Instrumentation & Controls System Installation Contractor. This term, ICS, I &C may all be used interchangeably. All work is to be performed per drawings and specifications.
- B. This Section, 17000 applies to all sections.

1.2 SCOPE

- A. I &C contractor shall provide instrumentation, equipment, installation, technical services, as shown on drawings, and/or Instrument Schedules, and /or Specifications. The I &C contractor shall coordinate specifications for the application and submit final instrument data sheets for approval by Micron (MTI) prior to procurement. Procurement of any item without the written approval from MTI is prohibited, and MTI shall have no obligation to pay for any items procured without approval.
- B. I &C contractor shall provide and install PLC's and PLC Remote Input / Output (RIO), panels, hardware, field devices. Contractor to provide systems integration unless otherwise noted in design documents.
- C. I &C contractor shall provide panels, power supplies, remote emergency shutdown stations, hardware, installation materials, so as to provide a complete Chemical Distribution Control system. MTI shall provide Chemical Distribution system programming.
- D. Telecom contractor shall provide (2) CAT 6 cables and pull string in a single conduit for ethernet runs between each PLC and telecom control room. One CAT 6 will be terminated and landed on CPU, the second CAT 6 will be left unterminated as a spare.
- E. I &C contractor shall provide panels, power supplies, hardware, installation materials, so as to provide a complete Chemical Distribution Control Tracetek leak detection system. MTI shall provide Chemical Distribution leak detection system programming.
- F. Provide for receipt and storage of Instrument Equipment. Receive, maintain inventory, and inspect equipment for damage and shortage. Prior to acceptance, verify equipment conforms to specified requirements. Report damaged equipment to Owner within five days of receipt or assume full responsibility for equipment replacement. Unload equipment using methods, which minimize damage. Store equipment in safe, dry location, protect from the elements, and in accordance with manufacturer recommendations. Deliver manuals, received with instruments to the Owner.
- G. Provide calibration and operational hardware bench checks of field instruments. This will prevent installation of defective instruments in the field. Provide certified

calibration documentation to the owner.

- H. Provide and install conduit, wire, raceways, and accessories to power and interconnect instrumentation and ICS equipment. Note areas marked Class 1, Div. 2.
- I. Provide and install pneumatic tubing, fittings, and accessories to power and interconnect pneumatic instrumentation.
- J. Provide and install control panels unless otherwise noted in design documents.
- K. Provide “red-marked” final P&IDs to reflect “as-built” conditions. P&IDs provided in this package shall be used as a base line. The I & C contractor shall operate in conjunction with MTI personnel to accomplish successful completion of project.
- L. Provide field checkout between the Field Termination Assembly Points and field instruments in order to verify that the instrument loop operates correctly (in the case of valves or actuators, verify correct direction per the Final Sequence of Operation).
- M. Provide for system startup of each Process, in a coordinated approved manner by MTI.
- N. Provide for generation of Final Acceptance Test procedure. Submit to MTI for review/acceptance of format prior to performing test.
- O. Provide for performance of Final Acceptance Test, witnessed by MTI for each Process.

1.3 QUALITY ASSURANCE

- A. Regulatory Requirements: Conform with requirements of National Electrical Code, National Fire Protection Association, and Local Codes.
- B. Standards: Comply with following requirements and recommendations wherever applicable.

1.4 SUBMITTALS

- A. Manufacturers Data as required in individual Sections. Submit substitutions for specified instrumentation for approval by M.T.I. prior to procurement.
- B. Provide for design and submittal of the following Data Items (five sets of electronic and hard copy required) I&C contractor shall follow naming rules, nomenclature for ICS according to MTI standards, providing the following:

1. Final "red-marked" P&IDs
2. Final Sequence of Operations
3. Final "red-marked" I/O database
4. Data sheets for field instruments (review with Owner prior to procurement)
5. Instrument Loop Wiring Diagrams
6. Instrument Wiring and Tubing Connection Drawings
7. Project Schedule (show P&ID/Sequence of Operation reviews, Simulation Demo Reviews, Data Item Submittal/Review Dates etc., order of process development to coincide with major Construction Milestones etc.)
8. Weekly Progress/Status Reports

1.5 DELIVERY, STORAGE, AND HANDLING OF PREPURCHASED EQUIPMENT

- A. Receive, maintain inventory, and inspect equipment for damage and shortage.
- B. Prior to acceptance, verify equipment conforms to specified requirements. Report damaged equipment to MTI within five days of receipt or assume full responsibility for equipment replacement.
- C. Unload equipment using methods which minimize damage.
- D. Store equipment in safe, dry location, protect from the elements, and in accordance with manufacturer recommendations.
- E. Deliver manuals, received with instruments to the MTI.

1.6 CONTRACTOR QUALIFICATIONS

- A. Demonstrate past semiconductor facility experience and capability to provide a complete and operational system to the satisfaction of MTI.

END OF SECTION 17000